

Scalable Optimal Transport for LEDH-PFPPF-OT: A Self-Contained Literature Survey

BayesFilter local literature survey

June 17, 2026

Abstract

LEDH-PFPPF-OT combines an invertible particle-flow proposal with an optimal transport or differentiable-resampling step. The scaling problem is not merely to compute an optimal-transport distance. The filter needs a transport object: a coupling, a barycentric projection, or a directly transported particle cloud that maps a weighted empirical ensemble to an approximately equal-weight ensemble. Dense entropic optimal transport provides such an object, but for N particles in state dimension D it requires all-pairs costs or kernels: $O(N^2D)$ arithmetic per full squared-Euclidean pass and $O(N^2)$ storage if the cost, kernel, or coupling is materialized. The current TensorFlow BayesFilter implementation already has a streaming mode that avoids storing the full transport matrix, but it still computes all row-column particle interactions.

This survey explains the LEDH-PFPPF-OT transport problem and reviews the main solution families in the scalable optimal-transport literature: exact online and GPU Sinkhorn methods, Nystrom kernel Sinkhorn, positive-feature Sinkhorn, direct low-rank coupling optimal transport, sparse and screened methods, accelerated Sinkhorn variants, stochastic and minibatch optimal transport, sliced and subspace optimal transport, and particle-filter-specific localization. The recommended research posture is a three-lane program: retain exact online Sinkhorn as the semantics-preserving reference, prototype factored entropic transports such as Nystrom and positive-feature Sinkhorn, and study low-rank coupling, localized/block, and sliced/subspace transports as explicit new resampling methods requiring downstream filtering validation.

1 Evidence Contract and Scope

Field	Contract
Question	How can the dense or all-pairs optimal-transport step in LEDH-PFPPF-OT be scaled for large particle count N and large state dimension D ?
Baseline	Current BayesFilter <code>annealed_transport_resample_tf</code> in <code>experiments/dpf_implementation/tf_tfp/resampling/annealed_transport_tf.py</code> .
Required object	A usable particle transport: a coupling, a barycentric projection, or a directly transported cloud.
Screen	Does the method reduce $N \times N$ memory or all-pairs $O(N^2 D)$ arithmetic while preserving or explicitly replacing the transport object?
Veto	A method that returns only a scalar loss; cannot handle weighted empirical measures; silently changes resampling semantics; or depends on a non-TensorFlow backend as the BayesFilter implementation default.
Not concluded	No method is default-ready from literature alone. No posterior-validity, gradient-correctness, HMC-readiness, or statistically supported ranking claim is made in this document.

The ResearchAssistant workspace at `/home/ubuntu/python/ResearchAssistant` was available through a read-only MCP adapter, but its local index returned no stored summaries for this topic. The evidence base is therefore the local PDF, text, and source corpus listed in Appendix A. A later arXiv download attempt for one additional 2025 source stalled and produced an incomplete file; that partial file is recorded in the manifest and is not used as evidence here.

2 The LEDH-PFPPF-OT Computation Problem

2.1 Particle-flow context

The Li–Coates PF-PF(LEDH) construction used in the local BayesFilter documentation is a proposal-corrected particle-flow particle filter [21]; the broader particle-flow filtering idea follows the Daum–Huang particle-flow line [9]. It is not merely a deterministic cloud motion. For particle i , the LEDH flow maintains both a zero-noise auxiliary path and an actual proposal path. With pseudo-time λ , the local observation linearization is evaluated at the moving auxiliary state:

$$H^i(\lambda) = \left. \frac{\partial h(\eta, 0)}{\partial \eta} \right|_{\eta=\bar{\eta}_{\lambda}^i}, \quad e^i(\lambda) = h(\bar{\eta}_{\lambda}^i, 0) - H^i(\lambda)\bar{\eta}_{\lambda}^i. \quad (1)$$

Given particle-specific predicted covariance P^i , the local LEDH coefficients are

$$A^i(\lambda) = -\frac{1}{2}P^i H^i(\lambda)^{\top} (\lambda H^i(\lambda) P^i H^i(\lambda)^{\top} + R)^{-1} H^i(\lambda), \quad (2)$$

$$b^i(\lambda) = (I + 2\lambda A^i(\lambda)) \left[(I + \lambda A^i(\lambda)) P^i H^i(\lambda)^{\top} R^{-1} \{z_k - e^i(\lambda)\} + A^i(\lambda) \bar{\eta}_0^i \right]. \quad (3)$$

On a pseudo-time grid $0 = \lambda_0 < \lambda_1 < \dots < \lambda_{N_{\lambda}} = 1$, the auxiliary and actual states are updated by the same local affine step:

$$\bar{\eta}_{\lambda_j}^i = \bar{\eta}_{\lambda_{j-1}}^i + \epsilon_j \left[A_j^i(\lambda_j) \bar{\eta}_{\lambda_{j-1}}^i + b_j^i(\lambda_j) \right], \quad (4)$$

$$\eta_{\lambda_j}^i = \eta_{\lambda_{j-1}}^i + \epsilon_j \left[A_j^i(\lambda_j) \eta_{\lambda_{j-1}}^i + b_j^i(\lambda_j) \right]. \quad (5)$$

The forward determinant multiplier is particle-specific:

$$\theta^i = \prod_{j=1}^{N_\lambda} \left| \det \{ I + \epsilon_j A_j^i(\lambda_j) \} \right|. \quad (6)$$

After the flow, Algorithm 1 uses a PF-PF importance update of the form

$$w_k^i \propto \frac{p(x_k^i | x_{k-1}^i) p(z_k | x_k^i) \theta^i}{p(\eta_0^i | x_{k-1}^i)} w_{k-1}^i. \quad (7)$$

If resampling is used, the documented Algorithm 1 state to resample is the triple

$$\{x_k^i, P_k^i, w_k^i\}_{i=1}^{N_p}. \quad (8)$$

Thus optimal-transport resampling is a BayesFilter extension layered on top of a proposal-corrected flow; the OT-resampling motivation is closer to ensemble transform and differentiable OT-resampling work [6, 28]. The survey question is how to make that extension scalable without confusing it with the Li-Coates source algorithm [21].

2.2 Why a scalar OT value is not enough

At a filtering time step, suppose the flow and likelihood update have produced particles $x_i \in \mathbb{R}^D$ with normalized weights a_i . A deterministic OT-resampling step compares the weighted empirical measure, in the spirit of ensemble-transform and differentiable OT resampling [6, 28],

$$\mu_N = \sum_{i=1}^N a_i \delta_{x_i} \quad (9)$$

to an equal-weight target ensemble. In the simplest self-transport version, the support points are the same cloud and the target weights are

$$b_j = \frac{1}{N}, \quad \nu_N = \sum_{j=1}^N b_j \delta_{x_j}. \quad (10)$$

A coupling P assigns source mass from particle i to target slot j . The transport polytope is

$$\Pi(a, b) = \{P \in \mathbb{R}_+^{N \times N} : P\mathbf{1} = a, P^\top \mathbf{1} = b\}. \quad (11)$$

The deterministic barycentric transform is

$$z_j = \frac{1}{b_j} \sum_{i=1}^N P_{ij} x_i. \quad (12)$$

If the implementation stores a row-stochastic target-to-source matrix T , the same transform is

$$z_i = \sum_{j=1}^N T_{ij} x_j, \quad \sum_j T_{ij} = 1. \quad (13)$$

This is the central computational requirement. A fast scalar OT(μ, ν) is insufficient for LEDH-PFOT. The filter needs P , a factorization of P , or an operator that can apply P to the particle

matrix X . This agrees with the ensemble transform particle filter, where random resampling is replaced by the deterministic linear transform [28]

$$x_j^a = \bar{x}_j^a = \sum_{i=1}^M p_{ij} x_i^f, \quad j = 1, \dots, M. \quad (14)$$

Differentiable OT resampling likewise recovers a regularized transport matrix from dual potentials [6]:

$$p_{\epsilon,i,j}^{OT} = a_i b_j \exp\{\epsilon^{-1}(f_i^* + g_j^* - c_{ij})\}. \quad (15)$$

2.3 The current BayesFilter bottleneck

The current TensorFlow implementation has dense and streaming transport modes. Dense mode materializes a transport matrix and applies $Z = TX$. Streaming mode returns an empty transport-matrix sentinel but still loops over row and column chunks. For each block it computes squared costs such as

$$C_{ij} = \frac{1}{2} \|\tilde{x}_i - \tilde{x}_j\|_2^2 \quad (16)$$

and block log-transport weights of the schematic form

$$\log T_{ij} = \frac{f_i + g_j - C_{ij}}{\epsilon} - \ell_j + \log N + \log w_j, \quad (17)$$

where ℓ_j is a column log normalizer. It then accumulates

$$z_i = \sum_j T_{ij} x_j. \quad (18)$$

Streaming therefore solves part of the memory problem, but not the all-pairs arithmetic problem. Each full Sinkhorn update or transport application still depends on all pairs of particles.

3 Discrete OT and Entropic Sinkhorn Basics

This section fixes notation for standard discrete OT and entropic Sinkhorn formulations [7, 27, 35].

Let

$$\mu = \sum_{i=1}^n a_i \delta_{x_i}, \quad \nu = \sum_{j=1}^m b_j \delta_{y_j}, \quad (19)$$

with $a \in \Delta_n$, $b \in \Delta_m$, and cost matrix

$$C_{ij} = c(x_i, y_j). \quad (20)$$

The Kantorovich problem is

$$\text{OT}(\mu, \nu) = \min_{P \in \Pi(a,b)} \langle C, P \rangle, \quad (21)$$

where

$$\Pi(a, b) = \{P \in \mathbb{R}_+^{n \times m} : P \mathbf{1}_m = a, P^\top \mathbf{1}_n = b\}. \quad (22)$$

Entropic regularization replaces the linear program by [7, 27]

$$\text{OT}_\epsilon(\mu, \nu) = \min_{P \in \Pi(a,b)} \langle C, P \rangle - \epsilon H(P), \quad (23)$$

with entropy convention

$$H(P) = - \sum_{ij} P_{ij} (\log P_{ij} - 1). \quad (24)$$

A relative-entropy version often used in differentiable resampling is

$$W_{2,\epsilon}^2(\alpha_N, \beta_N) = \min_{P \in S(a,b)} \sum_{i,j=1}^N \left[p_{ij} c_{ij} + \epsilon p_{ij} \log \frac{p_{ij}}{a_i b_j} \right]. \quad (25)$$

For $\epsilon > 0$, the minimizer is unique and has Gibbs scaling form

$$P_\epsilon = \text{diag}(u) K \text{diag}(v), \quad K = \exp(-C/\epsilon). \quad (26)$$

The basic Sinkhorn updates are

$$u \leftarrow a / (Kv), \quad v \leftarrow b / (K^\top u), \quad (27)$$

with division elementwise. Stabilized log-domain variants use dual potentials and log-sum-exp reductions rather than explicitly forming kernel entries that can underflow [34].

For particle transport, the coupling is applied as

$$Z = \text{diag}(b)^{-1} P^\top X, \quad (28)$$

or, under the row-stochastic orientation, as $Z = TX$. Every scalable method must explain how it computes this transported particle matrix.

4 Exact Online and GPU Sinkhorn

Online Sinkhorn preserves the exact entropic OT problem but avoids storing the full cost or kernel matrix, as in GeomLoss/KeOps-style reductions and recent GPU-specialized Sinkhorn solvers [5, 12, 36, 38]. For squared Euclidean costs,

$$C_{ij} = \|x_i - y_j\|_2^2 = \|x_i\|_2^2 + \|y_j\|_2^2 - 2x_i^\top y_j. \quad (29)$$

The stabilized update for one potential is a row-wise log-sum-exp reduction, schematically

$$f_i \leftarrow -\epsilon \log \sum_j \exp((g_j - C_{ij})/\epsilon + \log b_j), \quad (30)$$

and similarly for g_j . Online backends compute these reductions by streaming blocks of x_i, y_j rather than reading a stored C or K .

FlashSinkhorn observes that, for squared Euclidean costs, the log-domain update can be written as a biased dot-product log-sum-exp, close to transformer attention normalization [38]. It then uses tiled GPU kernels to stream data through on-chip memory. The paper is especially relevant because it also treats transport application through the barycentric projection

$$T_\epsilon(x_i) = \frac{1}{a_i} \sum_{j=1}^m P_{ij}^* y_j, \quad (31)$$

with the common gradient expression

$$\nabla_{x_i} \text{OT}_\epsilon = 2a_i(x_i - T_\epsilon(x_i)). \quad (32)$$

Thus exact online/GPU Sinkhorn is the semantics-preserving engineering lane: it improves memory traffic and GPU utilization while keeping the same entropic transport object. It does not, however, change the all-pairs asymptotic cost.

5 Nystrom Kernel Sinkhorn

The target paper *Massively scalable Sinkhorn distances via the Nystrom method* approximates the Gaussian Gibbs kernel [4]

$$K_{ij} = k_\eta(x_i, x_j) = \exp(-\eta\|x_i - x_j\|^2) \quad (33)$$

by a low-rank factorization

$$\tilde{K} = VA^{-1}V^\top. \quad (34)$$

Given landmark points

$$X_r = \{\tilde{x}_1, \dots, \tilde{x}_r\} \subset X, \quad (35)$$

the factors are

$$V_{ij} = k_\eta(x_i, \tilde{x}_j), \quad A_{jj'} = k_\eta(\tilde{x}_j, \tilde{x}_{j'}). \quad (36)$$

If $A = LL^\top$, a matrix-vector product is computed as

$$\tilde{K}v = V \left(L^{-\top} \left(L^{-1}(V^\top v) \right) \right). \quad (37)$$

After Cholesky factorization, each multiply costs $O(nr)$ rather than $O(n^2)$. Sinkhorn scaling uses

$$\tilde{P} = D_1 \tilde{K} D_2 = D_1 V A^{-1} V^\top D_2. \quad (38)$$

For particle transport, this factorization is directly useful:

$$\tilde{P}X = D_1 V A^{-1} V^\top D_2 X. \quad (39)$$

The adaptive Nystrom subroutine doubles the rank until the entrywise kernel approximation error is below a threshold. For Gaussian kernels, the diagnostic used in the paper is

$$\|K - \tilde{K}\|_\infty = 1 - \min_i \tilde{K}_{ii}. \quad (40)$$

If points lie in a ball in $\mathbb{R}^d$, the paper gives a worst-case bound of the form

$$r^*(X, \eta, \epsilon') \leq 3 \left(6 + \frac{53}{d} \eta R^2 + \frac{3}{d} \log \frac{2n}{\epsilon'} \right)^d. \quad (41)$$

This is not reassuring in high ambient dimension. The favorable case is low intrinsic dimension. If the data lie on a k -dimensional manifold, the bound becomes

$$r^*(X, \eta, \epsilon') \leq c_{\Omega, \eta} \left(\log \frac{n}{\epsilon'} \right)^{5k/2}. \quad (42)$$

The paper also states stability in logarithmic kernel error. If

$$\|\log K - \log \tilde{K}\|_\infty \leq \epsilon', \quad (43)$$

then scaling the approximate kernel gives a plan $\tilde{P} = D_1 \tilde{K} D_2$ with marginal residual

$$\|\tilde{P}\mathbf{1} - p\|_1 + \|\tilde{P}^\top \mathbf{1} - q\|_1 \leq \epsilon', \quad (44)$$

and controlled objective error. For LEDH-PFPP-OT, the essential extra diagnostic is transported-particle error,

$$\|\tilde{P}X - P_\epsilon X\|. \quad (45)$$

Nystrom Sinkhorn can help when the post-flow Gibbs kernel has small effective rank. It can fail when the effective rank is large, the entropy is too small, or high ambient dimension corresponds to high intrinsic dimension.

6 Positive-Feature Sinkhorn

Positive-feature Sinkhorn replaces the Gibbs kernel by an inner product of positive features [29]. Choose

$$\phi_\theta(x) \in \mathbb{R}_+^r, \quad (46)$$

define

$$k_\theta(x, y) = \langle \phi_\theta(x), \phi_\theta(y) \rangle, \quad (47)$$

and set the associated cost

$$c_\theta(x, y) = -\epsilon \log k_\theta(x, y). \quad (48)$$

For feature matrices

$$\xi = [\phi_\theta(x_1), \dots, \phi_\theta(x_n)] \in (\mathbb{R}_+^*)^{r \times n}, \quad (49)$$

$$\zeta = [\phi_\theta(y_1), \dots, \phi_\theta(y_m)] \in (\mathbb{R}_+^*)^{r \times m}, \quad (50)$$

the sample kernel is

$$K_\theta = \xi^\top \zeta. \quad (51)$$

Sinkhorn matrix-vector products therefore cost $O(r(n+m))$. The dual objective can be written

$$W_{\epsilon, c_\theta}(\mu, \nu) = \max_{\alpha \in \mathbb{R}^n, \beta \in \mathbb{R}^m} a^\top \alpha + b^\top \beta - \epsilon (\xi e^{\alpha/\epsilon})^\top \zeta e^{\beta/\epsilon} + \epsilon. \quad (52)$$

The paper also uses integral kernel representations,

$$k(x, y) = \int_U \phi(x, u)^\top \phi(y, u) d\rho(u), \quad (53)$$

and finite approximations

$$\phi_\theta(x) = \frac{1}{\sqrt{r}}(\phi(x, u_1), \dots, \phi(x, u_r)). \quad (54)$$

For particle resampling, the factored transport application is

$$PX = \text{diag}(u) \xi^\top \zeta \text{diag}(v) X. \quad (55)$$

The caution is semantic: a feature approximation changes the cost or kernel. The transported particle cloud and marginal residuals must be compared against the exact entropic reference.

7 Direct Low-Rank Coupling OT

Low-rank kernel methods approximate K . Direct low-rank OT constrains or parametrizes the coupling P itself [14, 30–32]. The nonnegative rank of a nonnegative matrix is

$$\text{rank}_+(M) = \min \left\{ r : M = \sum_{k=1}^r M_k, \text{rank}(M_k) = 1, M_k \geq 0 \right\}. \quad (56)$$

The rank-constrained coupling set is

$$\Pi_{a,b}(r) = \{P \in \Pi(a, b) : \text{rank}_+(P) \leq r\}. \quad (57)$$

The low-rank OT problem is

$$\min_{P \in \Pi_{a,b}(r)} \langle C, P \rangle. \quad (58)$$

The factored coupling parameterization is

$$P = Q \operatorname{diag}(1/g) R^\top, \quad (59)$$

where

$$Q \in \Pi(a, g), \quad R \in \Pi(b, g), \quad g \in \Delta_r. \quad (60)$$

Equivalently,

$$FC_{a,b}(r) = \{(Q, R, g) : Q \in \mathbb{R}_+^{n \times r}, R \in \mathbb{R}_+^{m \times r}, g \in (\mathbb{R}_+^*)^r, Q \in \Pi(a, g), R \in \Pi(b, g)\}. \quad (61)$$

Then

$$\min_{P \in \Pi_{a,b}(r)} \langle C, P \rangle = \min_{(Q, R, g) \in FC_{a,b}(r)} \langle C, Q \operatorname{diag}(1/g) R^\top \rangle. \quad (62)$$

Transport application is efficient:

$$PX = Q \operatorname{diag}(1/g) R^\top X. \quad (63)$$

A newer latent-coupling factorization introduces an additional $r \times r$ latent plan [17, 18]:

$$LC_{a,b}(r) = \{(Q, R, T) : Q \in \mathbb{R}_+^{n \times r}, R \in \mathbb{R}_+^{m \times r}, T \in \mathbb{R}_+^{r \times r}, Q \in \Pi(a, \cdot), R \in \Pi(b, \cdot), T \in \Pi(g_Q, g_R)\}. \quad (64)$$

The induced coupling is

$$P = Q \operatorname{diag}(1/g_Q) T \operatorname{diag}(1/g_R) R^\top. \quad (65)$$

This allows mass to route through a small latent transport plan rather than only through diagonal latent components.

The Riemannian low-rank line uses positive factors and manifold optimization [19]. For balanced linear OT, one representative parameterization is

$$\Gamma = U \operatorname{diag}(U^\top \mathbf{1})^{-1} V^\top, \quad (66)$$

with objective

$$f(U, V) = \langle C, \Gamma \rangle. \quad (67)$$

It also provides a rank-sufficiency diagnostic. Let Γ_r be the rank- r solution and define the reduced gradient or dual slack matrix

$$R = \nabla_\Gamma f(\Gamma_r) - \alpha^* \mathbf{1}^\top - \mathbf{1}(\beta^*)^\top. \quad (68)$$

Then

$$\delta_r = \min_{ij} R_{ij}. \quad (69)$$

Under the theorem's assumptions, $\delta_r \geq 0$ certifies global optimality for the full coupling problem; $\delta_r < 0$ identifies a direction in which rank should be increased. Even when the exact certificate is not achieved at a finite iterate, it is a useful diagnostic for whether the rank is too small.

Direct low-rank coupling is a promising subquadratic route because it reduces storage and transport application. It is also a real algorithmic change: it solves over low-rank couplings, not the original dense entropic feasible set.

8 Sparse, Screened, and Stabilized OT

Sparse multiscale OT begins with the exact discrete problem [33]

$$\Pi(\mu, \nu) = \{\pi \in P(X \times Y) : \pi(\{x\} \times Y) = \mu(x), \pi(X \times \{y\}) = \nu(y)\}, \quad (70)$$

$$\min_{\pi \in \Pi(\mu, \nu)} C(\pi), \quad C(\pi) = \sum_{(x, y) \in X \times Y} c(x, y) \pi(x, y). \quad (71)$$

The dense problem considers the full product $X \times Y$. Sparse multiscale methods solve a sequence of sparse subproblems while verifying global optimality through local shielding neighborhoods. This is relevant if post-flow LEDH particles have genuinely local couplings.

Screenhorn accelerates regularized OT by screening dual variables [2]. The regularized primal is

$$S_\eta(\mu, \nu) = \min_{P \in \Pi(\mu, \nu)} \{\langle C, P \rangle - \eta H(P)\}. \quad (72)$$

With Gibbs kernel

$$K = \exp(-C/\eta), \quad (73)$$

the dual can be written

$$\min_{u \in \mathbb{R}^n, v \in \mathbb{R}^m} \Psi(u, v) = \mathbf{1}_n^\top B(u, v) \mathbf{1}_m - \langle u, \mu \rangle - \langle v, \nu \rangle, \quad (74)$$

where

$$B(u, v) = \Delta(e^u) K \Delta(e^v). \quad (75)$$

The idea is to identify negligible components and solve a smaller constrained dual problem. As with sparse multiscale OT, the method is useful only if the screened plan still gives acceptable transport and marginal residuals.

Stabilized sparse scaling, epsilon scaling, truncation, and coarse-to-fine methods should be treated as a toolbox rather than a single replacement algorithm [34]. They are particularly important when ϵ is small and standard-domain $K = \exp(-C/\epsilon)$ becomes numerically fragile.

9 Accelerated, Greedy, and Stochastic Sinkhorn Iterations

This family improves iteration complexity or update scheduling but does not by itself remove dense kernel access [1, 3, 11, 22, 37]. Greenhorn updates the most violating row or column rather than all rows and columns. The plan still has the scaling form

$$P = \text{diag}(u) K \text{diag}(v), \quad (76)$$

but the update schedule for components of u and v changes. Accelerated primal-dual, accelerated mirror-descent, and related methods improve theoretical dependence on accuracy or regularization. If each iteration still requires dense Kv and $K^\top u$, the large N, D bottleneck remains. These methods are secondary tools to combine with online, sparse, or low-rank backends.

10 Stochastic and Minibatch OT

Stochastic OT methods are important for learning losses and cases where the measures are observed through samples [15, 16]. Particle filtering is different: at a given time step the resampling transform acts on a finite weighted ensemble. Replacing the full coupling by stochastic dual estimates is not automatically a valid resampling map.

Minibatch OT methods reduce computation by solving smaller OT problems and an outer problem over minibatches [23]. Entropic population BoMb-OT has the form

$$ED_d^m(\mu, \nu) = \min_{\gamma \in \Pi(\mu^{\otimes m}, \nu^{\otimes m})} \mathbb{E}_{(X^m, Y^m) \sim \gamma} [d(P_{X^m}, P_{Y^m})] + \lambda \text{KL}(\gamma \mid \mu^{\otimes m} \otimes \nu^{\otimes m}). \quad (77)$$

For finite k minibatches of size m , an outer $k \times k$ OT problem couples minibatches. If the inner discrepancy is Sinkhorn, cost construction is roughly $O(m^2 k^2)$. If the inner discrepancy is sliced Wasserstein, it can be closer to $O(m \log(m) k^2)$.

An unbalanced minibatch variant uses local unbalanced OT [24]:

$$UOT_\tau^\phi(\mu_n, \nu_n) = \min_{\pi \in \mathbb{R}_+^{n \times n}} \langle C, \pi \rangle + \tau D_\phi(\pi \mathbf{1}, \mu_n) + \tau D_\phi(\pi^\top \mathbf{1}, \nu_n). \quad (78)$$

Minibatch OT is tempting for large N , but it is a dangerous drop-in replacement. A minibatch plan can be optimal inside batches and wrong for the full ensemble. It should be treated as a new randomized or approximate resampling method, not as a faithful acceleration of full-ensemble transport.

11 Sliced, Max-Sliced, Generalized Sliced, and Subspace OT

Sliced methods attack high dimension by projecting measures to one-dimensional distributions [10, 20, 25, 26]. For one-dimensional measures, the monotone transport map is

$$T(x) = F_\nu^{-1}(F_\mu(x)), \quad (79)$$

and

$$W_p^p(\mu, \nu) = \int_0^1 |F_\mu^{-1}(z) - F_\nu^{-1}(z)|^p dz. \quad (80)$$

For empirical distributions this is computed by sorting projected samples.

For $\theta \in S^{d-1}$, let $\theta_\# \mu$ be the pushforward of μ under $x \mapsto \langle \theta, x \rangle$. The sliced Wasserstein distance is

$$\text{SW}_p^p(\mu, \nu) = \int_{S^{d-1}} W_p^p(\theta_\# \mu, \theta_\# \nu) d\theta. \quad (81)$$

The empirical approximation uses directions $\theta_1, \dots, \theta_L$:

$$\widehat{\text{SW}}_p^p(\mu, \nu) = \frac{1}{L} \sum_{\ell=1}^L W_p^p((\theta_\ell)_\# \mu, (\theta_\ell)_\# \nu). \quad (82)$$

Equivalently, for distributions P_d, P_g ,

$$\widetilde{W}_p(P_d, P_g) = \left[\int_{\omega \in \Omega} W_p^p(P_d^\omega, P_g^\omega) d\omega \right]^{1/p}. \quad (83)$$

For finite samples D, F and directions $\widehat{\Omega}$,

$$\frac{1}{|\widehat{\Omega}|} \sum_{\omega \in \widehat{\Omega}} W_2^2(D^\omega, F^\omega) \quad (84)$$

with each one-dimensional term computed by sorting,

$$W_2^2(D^\omega, F^\omega) = \frac{1}{|D|} \sum_i \left\| D_{\pi_D^\omega(i)}^\omega - F_{\pi_F^\omega(i)}^\omega \right\|_2^2. \quad (85)$$

Max-sliced Wasserstein replaces random averaging by a best projection:

$$\text{MSW}_p^p(\mu, \nu) = \max_{\theta \in S^{d-1}} W_p^p(\theta_\# \mu, \theta_\# \nu). \quad (86)$$

Sliced and subspace methods are highly relevant when D is the dominant problem. They can generate projected transport maps, but a full particle resampling transform in $\mathbb{R}^D$ is not the same object as a dense coupling. They should be studied as new resampling approximations unless a precise construction maps the projected couplings back to a validated full-state transport.

12 Localization and Particle-Filter Structure

The generic OT literature does not know the state-space model. LEDH-PFPP-OT does. If the state has spatial, block, factor, or conditional-independence structure, localization may be more principled than a generic global OT approximation. A localized OT resampler could partition the state as

$$x_i = (x_i^{(1)}, \dots, x_i^{(B)}), \quad (87)$$

then compute block transports

$$P^{(b)} \in \Pi(a^{(b)}, b^{(b)}) \quad (88)$$

or use shared particle-level couplings with local costs

$$C_{ij} = \sum_{b=1}^B C_{ij}^{(b)}. \quad (89)$$

This lane should be studied alongside low-rank methods, especially for spatial fields, macro panels, or structured latent states.

13 Method Taxonomy

Family	Representative sources	Transport object?	What it fixes	Main risk
Exact online/GPU Sinkhorn	[5, 12, 36, 38]	Yes	Dense storage and GPU IO	Still all-pairs $O(N^2 D)$ work
Nystrom Sinkhorn	[4]	Yes, factored entropic coupling	Kernel matvecs when effective rank is small	Rank can grow badly with dimension or small ϵ

Family	Representative sources	Transport object?	What it fixes	Main risk
Positive-feature Sinkhorn	[29]	Yes if coupling application is implemented	Linear-time positive kernel matvecs	Feature approximation changes cost/kernel
Low-rank coupling OT	[14, 17–19, 30–32]	Yes, factored coupling	Coupling storage and transport application	Changes feasible plan class
Sparse/multiscale/screenable	[2, 33, 34]	Yes	Active pair count when coupling is local or screenable	Locality may fail in high dimension
Accelerated/greedy Sinkhorn	[1, 3, 11, 22, 37]	Yes	Iteration count/convergence	Dense kernel access remains
Stochastic/minibatch OT	[15, 16, 23, 24]	Sometimes partial/local plans	Training-scale memory and compute	Not a full deterministic resampling plan
Sliced/subspace OT	[10, 20, 25, 26]	Projected plans/maps	High-dimensional statistics and speed	Different transport object
Localization/block OT	[28] and model structure	Yes if designed	Large D with state locality	Requires model-specific validation

14 Code Availability and Reuse Risk

Source availability is a key practical filter, but it should not be confused with default-readiness. In this section, a *valid inspected checkout* means that the repository is locally available under `.localsource/scalable_ot_code_audit`, has a readable Git commit, and the listed implementation files were inspected. It does not mean that the package was installed, built, or tested. Because BayesFilter-owned algorithmic code defaults to TensorFlow/TFP, PyTorch, JAX, Triton, and C++ repositories below are treated as algorithmic references unless a later plan explicitly approves a backend exception.

Repository/method	Local audit status	Backend	Transport object exposed?	BayesFilter decision implication
FilterFlow ferentiable [6]	dif- <code>filterflow</code> , PF mit <code>5d8300b</code> ; inspected checkout	com- TensorFlow/TFP	Yes. <code>plan.py</code> exposes <code>transport_from_potentials</code> and <code>transport</code> , returning a dense transport matrix from <code>sinkhorn.py</code> potentials.	Most directly compatible source for differentiable and resampling semantics, but not a scalable fix because it materializes dense $[B, N, N]$ transport.
POT / Python Optimal Transport [13]	POT, commit <code>3e10dd6</code> ; valid inspected checkout	Multi-backend: NumPy, PyTorch, JAX, TensorFlow, CuPy	Yes. Dense Sinkhorn, Greenkhorn, Screenkhorn, Nystrom lazy plans, low-rank Sinkhorn, factored OT, and sliced plans are present in <code>ot/bregman</code> , <code>ot/lowrank.py</code> , <code>ot/factored.py</code> , and <code>ot/sliced</code> .	Best broad protocol to inform TensorFlow work, but BayesFilter still needs local wrappers and resampling-specific validation.
OTT-JAX [8]	<code>ott-sparse</code> , commit <code>e241ec3</code> ; valid inspected sparse checkout	JAX	Yes. <code>SinkhornOutput</code> and <code>LRSinkhornOutput</code> expose <code>matrix</code> , <code>apply</code> , and <code>point-cloud</code> support online/factored applications.	Excellent API reference for exact online and low-rank transport operators, but not a default BayesFilter backend.

Repository/method	Local audit status	Backend	Transport object exposed?	BayesFilter decision implication
GeomLoss KeOps Sinkhorn [5, 12]	/ <code>geomloss</code> , online <code>00e493f</code> ; valid checkout	commit PyTorch/KeOps	Yes. The result API exposes <code>plan</code> , <code>lazy_plan</code> , and <code>plan_operator</code> ; point-cloud implementations expose lazy transport-plan pose densities.	Strong exact online Sinkhorn reference for avoiding dense storage. It still performs all-pairs interactions and would require a TensorFlow port or backend exception.
FlashSinkhorn [38]	<code>flash-sinkhorn</code> , com- mit <code>d538da3</code> ; valid in- spected checkout	PyTorch/Triton	Yes. The API documents <code>apply_plan_vec</code> , <code>flashstyle</code> and <code>apply_plan_mat_flashstyle</code> , but solver code returns shifted potentials for GPU plan application.	Best source for IO-aware GPU kernel design. High porting risk into TensorFlow, but valuable for a future fused exact-reference implementation.
LinearSinkhorn / positive-feature and Nystrom scripts [4, 29]	<code>LinearSinkhorn</code> , com- mit <code>01943c7</code> ; valid in- spected checkout	PyTorch	Partial. <code>FastSinkhorn.py</code> implements <code>Lin_Sinkhorn</code> , and <code>Nys_Sinkhorn</code> , and feature/Nystrom helpers, but reusable particle transport application is not exposed as a package API.	Useful for deriving positive-feature/Nystrom prototypes; not implementation-ready for BayesFilter.
Schmitzer sparse/multiscale code [33, 34]	<code>MultiScaleOT</code> commit <code>48a47aa</code> ; <code>optimal-transport</code> commit <code>1d1aaf4</code> ; valid inspected checkouts	C++ with Python tooling	Yes, through sparse coupling handlers, short-cut/shielding solvers and stabilized sparse Sinkhorn source under <code>src/ShortCutSolver</code> and <code>src/Sinkhorn</code> .	Strong sparse/multiscale reference if post-flow particles have exploitable locality. High integration risk and likely unsuitable as first TensorFlow prototype.
Mini-batch-OT BoMb-OT [23, 24]	/ <code>Mini-batch-OT-sparse</code> , commit <code>1ef5ffc</code> ; par- tial checkout after DNS/GnuTLS/HTTP2 failures	NumPy/POT scripts	Partially visible. <code>ABC/utls.py</code> and <code>ColorTransfer/utls.py</code> compute local mini-batch stochastic minibatch semiplans and higher-level couplings, but the checkout is dirty/incomplete.	Keep as concept/literature lane until a clean archive is available. Even with source, semantics do not directly give a single full deterministic resampling plan.

The practical ranking suggested by the source audit is therefore different from a pure literature ranking. For a near-term BayesFilter prototype, FilterFlow is the best source-faithful TensorFlow reference, POT and OTT-JAX are the most useful factored/low-rank implementation references, GeomLoss and FlashSinkhorn are the best exact online/GPU design references, and Schmitzer’s code is a specialized sparse/multiscale reference. Mini-batch-OT should not be used as a determining source-code factor until the checkout is clean or the user provides an archive.

15 Recommended Research Program

Lane A: exact reference. Keep the current dense/streaming TensorFlow path as the correctness reference. Study FlashSinkhorn or KeOps-style online GPU design as an implementation benchmark [5, 36, 38]. This lane answers how far exact entropic OT scales if memory traffic and kernel fusion are fixed. Required diagnostics include

$$\|P\mathbf{1} - a\|_1, \quad \|P^\top \mathbf{1} - b\|_1, \quad \|Z_{\text{candidate}} - Z_{\text{dense}}\|. \quad (90)$$

Lane B: approximate entropic factored kernels. Prototype fixed-rank Nystrom Sinkhorn first, then compare positive-feature Sinkhorn [4, 29]. Both should expose a factored transport application,

$$Z = \text{diag}(b)^{-1} \tilde{P}^\top X, \quad (91)$$

without dense P . Primary diagnostics are effective rank, kernel approximation error, marginal residual, and transported-particle error.

Lane C: structural approximate couplings. Study low-rank coupling OT, localized OT, and sliced/subspace transports as new resampling methods [14, 25, 26, 30, 31]. These methods should not be judged only by dense Sinkhorn parity, because some may intentionally bias the transport to improve high-dimensional behavior. They require downstream filtering diagnostics.

16 Validation Ladder Before Any Default Change

1. Small deterministic dense parity: compare transported particles, not only OT values.
2. Marginal checks: source and target constraints with weighted inputs.
3. Positivity and non-finite checks.
4. Representation diagnostics: rank, feature dimension, sparsity, projection count, or block locality.
5. LEDH-PFPPF-OT value parity on existing fixtures.
6. Gradient smoke tests after value parity.
7. Runtime and memory ladder over N, D .
8. Posterior/reference diagnostics for filtering.
9. Multi-seed uncertainty-aware comparisons before ranking stochastic methods.
10. Default-readiness review only after downstream evidence, not from paper benchmarks alone.

17 Conclusions

The answer to whether the Nystrom Sinkhorn paper 1812.05189 can help is yes, but it should not be the only method considered [4]. Nystrom Sinkhorn is a strong first approximate entropic transport because it yields a factored coupling and can apply the transport to particles. Its success depends on the effective rank of the post-flow Gibbs kernel.

For near-term engineering, exact online/GPU Sinkhorn is the reference lane. It preserves the current entropic OT semantics and directly supports barycentric transport, but it remains all-pairs. For true large- N, D scaling, the most promising methods are structural: low-rank coupling OT, localization/block OT, and sliced/subspace transports. These are not mere accelerations; they are new resampling approximations and must be validated as such.

The safest research posture is a three-lane program: exact online Sinkhorn for reference and moderate scaling, low-rank kernel Sinkhorn for the first factored entropic approximation, and low-rank/localized/sliced transports as explicit new filtering methods. No candidate should become a BayesFilter default until it passes transported-particle parity or a declared replacement criterion, marginal diagnostics, numerical stability checks, and downstream filtering validation.

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A Local Source Map

Topic	Local source
Corpus manifest	<code>.localsource/scalable_ot_survey/MANIFEST.md</code>
Code audit manifest	<code>.localsource/scalable_ot_code_audit/MANIFEST.md</code>
Current TensorFlow annealed transport	<code>experiments/dpf_implementation/tf_tfp/resampling/annealed_transport_tf.py</code>
LEDH/PF-PF source-form documentation	<code>docs/chapters/ch19b_dpf_literature_survey.tex</code> ; <code>docs/chapters/ch19c_dpf_implementation_literature.tex</code>
Cuturi, Sinkhorn distances	<code>.localsource/dpf_ot_audit/cuturi2013_sinkhorn_distances.txt</code>
Reich, ensemble transform particle filter	<code>.localsource/dpf_ot_audit/reich2013_nonparametric_ensemble_transform.txt</code>
Corenflos et al., differentiable OT resampling	<code>.localsource/dpf_ot_audit/corenflos2021_differentiable_particle_filtering_eot.txt</code>
Nystrom Sinkhorn target paper	<code>.localsource/1812.05189-src/sections/nystrom.tex</code> ; <code>.localsource/1812.05189-src/sections/sinkhorn.tex</code> ; <code>.localsource/1812.05189.txt</code>
Positive-feature Sinkhorn	<code>.localsource/scalable_ot_survey/2006.07057.txt</code>
Exact online/GPU Sinkhorn	<code>.localsource/scalable_ot_survey/1810.08278.txt</code> ; <code>.localsource/scalable_ot_survey/2004.11127.txt</code> ; <code>.localsource/scalable_ot_survey/2602.03067.txt</code> ; <code>.localsource/scalable_ot_survey/2605.00837.txt</code>

Topic	Local source
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Sparse/screened/stabilized OT	.localsource/scalable_ot_survey/1510.05466.txt; .localsource/scalable_ot_survey/1906.08540.txt; .localsource/dpf_ot_audit/schmitzer2019_stabilized_sparse_scaling.txt
Accelerated/greedy Sinkhorn	.localsource/scalable_ot_survey/1705.09634.txt; .localsource/scalable_ot_survey/1802.04367.txt; .localsource/scalable_ot_survey/1803.01347.txt; .localsource/scalable_ot_survey/1901.06482.txt; .localsource/scalable_ot_survey/2605.30267.txt
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Sliced/subspace OT	.localsource/scalable_ot_survey/1901.08949.txt; .localsource/scalable_ot_survey/1902.00434.txt; .localsource/scalable_ot_survey/1904.05877.txt; .localsource/scalable_ot_survey/2508.12519.txt
Inspected scalable OT code repositories	.localsource/scalable_ot_code_audit/filterflow; .localsource/scalable_ot_code_audit/POT; .localsource/scalable_ot_code_audit/ott-sparse; .localsource/scalable_ot_code_audit/geomloss; .localsource/scalable_ot_code_audit/flash-sinkhorn; .localsource/scalable_ot_code_audit/LinearSinkhorn; .localsource/scalable_ot_code_audit/schmitzer-MultiScaleOT; .localsource/scalable_ot_code_audit/schmitzer-optimal-transport; .localsource/scalable_ot_code_audit/Mini-batch-OT-sparse

B Incomplete Download Note

An additional attempt to download a recent arXiv source produced the partial file `.localsource/scalable_ot_survey/2505.06835`. The file has a PDF header but fails PDF metadata inspection because the trailer/xref table is missing. It is therefore not treated as an inspected source in this survey.